

```
plt.style.use('seaborn')
fig, ax = plt.subplots()
❷ ax.scatter(x_values, y_values, s=10)

# Set chart title and label axes.
--snip--

# Set the range for each axis.
❸ ax.axis([0, 1100, 0, 1_100_000])

plt.show()
```

We start with a range of x -values containing the numbers 1 through 1,000 ❶. Next, a list comprehension generates the y -values by looping through the x -values (for x in x_values), squaring each number ($x**2$), and assigning the results to y_values . We then pass the input and output lists to `scatter()` ❷. Because this is a large dataset, we use a smaller point size.

Before showing the plot, we use the `axis()` method to specify the range of each axis ❸. The `axis()` method requires four values: the minimum and maximum values for the x -axis and the y -axis. Here, we run the x -axis from 0 to 1,100 and the y -axis from 0 to 1,100,000. Figure 15-7 shows the result.

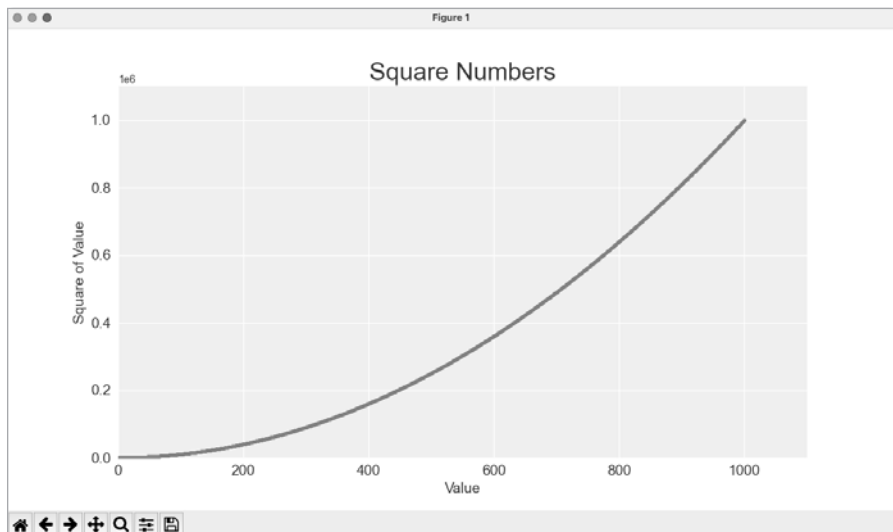


Figure 15-7: Python can plot 1,000 points as easily as it plots 5 points.

Customizing Tick Labels

When the numbers on an axis get large enough, Matplotlib defaults to scientific notation for tick labels. This is usually a good thing, because larger numbers in plain notation take up a lot of unnecessary space on a visualization.